

Sustainability at Scale

SEBI BRSR mandates 130+ ESG indicators. Net-zero by 2070. Climate AI is not NGO work anymore — it's a ₹25,000 cr compliance + opportunity market.

Does your employer have an ESG officer? What tools do they use?

WHAT'S INSIDE

- 01 4 applications
- 02 Satellite environmental monitoring
- 03 ESG reporting automation
- 04 Grid + renewable
- 05 Analogy

WHY THIS LESSON MATTERS

1,000 Indian companies must now report ESG. Most still do it manually.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

4 Indian climate AI use-cases.

2

Satellite + ML for monitoring.

3

ESG reporting automation.

4

Grid + renewable optimisation.

5

Carbon-accounting system for SME.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

4 applications

Key concept of this lesson.

02

**Satellite
environmental...**

Free + powerful

03

**ESG reporting
automation**

BRSR requires 130+ indicators

04

Grid + renewable

Discom-side AI

01

PART 1 OF 4

4 applications

Key concept of this lesson.

CONCEPT 1 OF 4

4 applications

A core idea you'll use throughout this lesson.

DEFINITION

4 applications

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Satellite environmental monitoring

Sentinel-5P: methane + NO₂ daily.

Sentinel-2 + CNN: illegal deforestation < 1 ha.

Time-series anomaly: oil spills.

Crop-residue burning: northern India post-harvest.

All via Google Earth Engine — free.

STACK

Sentinel + GEE

District-level daily.

KEY POINTS

- Sentinel-5P: methane + NO₂ daily
- Sentinel-2 + CNN: illegal deforestation < 1 ha
- Time-series anomaly: oil spills
- Crop-residue burning: northern India post-harvest

CONCEPT 3 OF 4

ESG reporting automation

Daily ingest of reports + policies.

LLM classifies + extracts indicators.

Flag inconsistencies.

Generate draft sections.

Saves 400+ hours per company per cycle.

MARKET

1,000+ cos

All need automation.

DEFINITION

ESG reporting automation

BRSR requires 130+ indicators

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Grid + renewable

Solar + wind 6-hr forecasts.

Demand per feeder.

Battery dispatch.

Predictive maintenance for transmission.

Fast-growing Indian opportunity.

GROWTH

Discom AI

Net-zero demands it.

KEY POINTS

- Solar + wind 6-hr forecasts
- Demand per feeder
- Battery dispatch
- Predictive maintenance for transmission

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where will you apply this in your next project?

Q2

What was the single biggest 'aha' from this lesson?

Q3

Which habit will you start using this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

BRSR covers:



All companies



Top-1,000 listed



Startups



Public only

BRSR covers:



CORRECT ANSWER

Top-1,000 listed

WHY

Mandatory from 2023

Sentinel-5P tracks:

A

Crops

B

Methane + NO₂

C

Only water

D

Random

Sentinel-5P tracks:



CORRECT ANSWER

Methane + NO₂

WHY

Daily atmospheric

ESG automation saves:

A

1 hr

B

400+ hrs/cycle

C

10 hrs

D

None

ESG automation saves:



CORRECT ANSWER

400+ hrs/cycle

WHY

Per company per cycle

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1

4 applications

2

Satellite environmental...

3

ESG reporting automation

4

Grid + renewable

Sustainability = market.

"India's net-zero promise is an AI market. Build for it."

DELIVERABLES

- NEXT
- Projects — industry pilot + pitch.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 9

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.